a

Sequential WSI Task Stream

Task 1 — *NSCLC*
- LUAD n=536
- LUSC n=509

Task 2 — *BRCA*
- IDC n=805
- ILC n=204

Task 3 — *RCC*
- CCRCC n=517
- PRCC n=296
- CHRCC n=121

Task 4 — *ESCA*
- Adenocarcinoma n=61
- Squamous cell carcinoma n=87

Task 5 — *TGCT*
- Seminoma n=180
- Mixed germ cell tumor n=67

Task 6 — *CESC*
- Adenocarcinoma n=46
- Squamous cell carcinoma n=225

b

Discover: Offline Semantic Top-512 Region Selection

Class descriptions

LUAD
- adenocarcinoma
- lung adenocarcinoma
- adenocarcinoma of the lung

LUSC ...

Prompt templates

Diagnosis: [CLASS]
A histopathology image of [CLASS]
An H&E-stained image of [CLASS]
...

TEXT Encoder (CONCH v1) → Class prototypes

Whole-slide image → Segment / Crop → ~2000 Regions

IMAGE Encoder (CONCH v1) → Region features

Semantic Scoring: Low relevance … High relevance

Top-K Selection → Region relevance scores

Selected regions (up to 512 per WSI)

c

Condense & Consolidate: Online Memory-Guided Continual Learning

8 × 8 patch features

Condense: RaRP-Fuse
- Q → Patch-to-Region Cross-Attention
- K, V
- RaRP-Compress

Patch-to-region update

Consolidate: Region-MAG Memory Anchor Guidance
- Memory anchor Guidance
- Old route tokens — KL — Current route tokens

Old-to-current update

Consolidate: Slide-MAG → Slide tokens → Prediction

Feature replay

Anchor replay

Replay buffer → Old patch tokens → RaRP-Compress → Memory route anchors → L2 Norm → **Old route relation** (cached until buffer update)

Trainable

Frozen